



SENTIMENT ANALYSIS ON GAME CHATS

**"STEP INTO A NEW WORLD WHERE ADVENTURE
AWAITS—PLAY BOLDLY, GAME SAFELY!"**



AGENDA

- **ABSTRACT**
- **INTRODUCTION**
- **PROBLEM STATEMENT**
- **LITERATURE REVIEW**
- **DATA COLLECTION**
- **METHODOLOGY**
- **RESULT**
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ABSTRACT



THIS PROJECT ADDRESSES THE GROWING ISSUE OF TOXIC BEHAVIOR IN ONLINE GAMING COMMUNITIES, SPECIFICALLY IN VALORANT AND MINECRAFT. IT AIMS TO DEVELOP AN AUTOMATED SYSTEM TO CLASSIFY IN-GAME CHAT MESSAGES AS TOXIC OR NEUTRAL USING NATURAL LANGUAGE PROCESSING (NLP). THE DATA IS PRIMARY, COLLECTED AS SCREENSHOTS, WITH TEXT EXTRACTED USING OCR TOOLS. THE SYSTEM'S NOVELTY LIES IN ITS UNIQUE COMBINATION OF IMAGE-BASED DATA COLLECTION AND MULTIMODAL ANALYSIS, ENSURING DIVERSE VOCABULARY DETECTION WHILE ADDRESSING REAL-TIME MODERATION CHALLENGES. BY LEVERAGING ADVANCED DEEP LEARNING TECHNIQUES, THIS PROJECT PROVIDES A SOLUTION TO ENHANCE PLAYER INTERACTIONS AND FOSTER A SAFER, MORE INCLUSIVE GAMING ENVIRONMENT.

INTRODUCTION



GAMING HAS EVOLVED DRAMATICALLY SINCE THE 1970s, FROM SIMPLE GAMES LIKE PAC-MAN TO TODAY'S IMMERSIVE MULTIPLAYER EXPERIENCES LIKE VALORANT AND MINECRAFT. THE RISE OF ONLINE GAMING IN THE LATE 1990s TRANSFORMED GAMING INTO A SOCIAL ACTIVITY, CONNECTING MILLIONS WORLDWIDE. HOWEVER, THIS GROWTH ALSO BROUGHT CHALLENGES, SUCH AS TOXIC BEHAVIOR IN IN-GAME CHATS, WHICH HARM PLAYER EXPERIENCES AND COMMUNITY INCLUSIVITY. TO ADDRESS THIS, DEVELOPERS HAVE TURNED TO AI AND NLP FOR AUTOMATED TOXICITY MODERATION. THIS PROJECT LEVERAGES A CUTTING-EDGE NLP MODEL TO CLASSIFY IN-GAME CHATS AS TOXIC OR NEUTRAL, AIMING TO CREATE A SAFER AND MORE POSITIVE GAMING ENVIRONMENT FOR ALL PLAYERS.

PROBLEM STATEMENT



TOXIC BEHAVIOR IN ONLINE GAMING, ESPECIALLY IN COMPETITIVE TITLES LIKE VALORANT, REMAINS A PERSISTENT ISSUE DESPITE RIOT GAMES AND MOJANG STUDIOS MODERATION EFFORTS. WHILE THE COMPANY HAS ISSUED NUMEROUS CHAT RESTRICTIONS AND BANS, MANY PLAYERS REPORT THAT TOXICITY, INCLUDING HARASSMENT AND VERBAL ABUSE, CONTINUES LARGELY UNCHECKED. FOR INSTANCE, IN JANUARY 2022, RIOT GAMES ISSUED OVER 400,000 TEXT AND VOICE CHAT RESTRICTIONS AND BANNED MORE THAN 40,000 PLAYERS FOR TOXIC COMMUNICATIONS. COMPLAINTS ON PLATFORMS LIKE REDDIT HIGHLIGHT THAT REPORTING SYSTEMS OFTEN FEEL INEFFECTIVE, WITH LITTLE VISIBLE ACTION TAKEN AGAINST REPEAT OFFENDERS. ADDITIONALLY, STUDIES SHOW THAT OFFENSIVE BEHAVIOR REMAINS WIDESPREAD, RAISING CONCERNS ABOUT THE LIMITATIONS OF CURRENT MODERATION TOOLS.



r/VALORANT • 4 yr. ago

HotClerk



Haves Reported Dozens of Toxic Players None Of Them Banned

Discussion

Hi,

I have reported dozens of players for toxic behavior in both text and voice comms. I have yet to be notified that there has been action taken for a single one of the reports. I understand a ban can be harsh but I haven't been notified of any time of action at all.

I am a girl and it can be discouraging to communicate with my team. These are valid reports that would fall under what Riot considers as toxic behavior. Can someone give me some insight about their experience?

A REDDIT POST BY A PLAYER EXPRESSING UNSATISFACTION WITH MODERATION METHODS

LITERATURE REVIEW



“ PREVIOUS STUDIES SHOW THAT TAILORED LEXICONS AND MACHINE LEARNING MODELS IMPROVE TOXICITY DETECTION IN GAME CHATS. RESEARCH HAS ADDRESSED CONTEXT-SPECIFIC LANGUAGE AND GENDER-BASED TOXICITY. THIS PROJECT BUILDS ON THESE FINDINGS USING A BERT MODEL TO CLASSIFY VALORANT AND MINECRAFT CHATS, ENHANCING MODERATION AND INCLUSIVITY “

SENTIMENT ANALYSIS IN DOTA 2

SINGH’S RESEARCH (2022) FOCUSES ON ANALYZING IN-GAME CHAT DATA FROM DOTA 2 TO IDENTIFY INSTANCES OF CYBERBULLYING. RECOGNIZING THAT TRADITIONAL SENTIMENT ANALYSIS TOOLS MAY NOT EFFECTIVELY CAPTURE THE UNIQUE LANGUAGE USED IN GAMING COMMUNITIES, SINGH DEVELOPED AN UPDATED LEXICON TAILORED TO DOTA 2’S SPECIFIC TERMINOLOGY. THIS APPROACH AIMED TO ENHANCE THE ACCURACY OF SENTIMENT CLASSIFICATION WITHIN THE GAME’S CONTEXT. HIGHLY NEGATIVE LOGS WERE RECOMMENDED FOR FURTHER REVIEW BY HUMAN MODERATORS TO ENSURE APPROPRIATE HANDLING OF POTENTIAL CYBERBULLYING INCIDENTS.



SENTIMENT ANALYSIS IN STARCRAFT 2

THOMPSON ET AL.(2017) EXTENDED A LEXICON-BASED SENTIMENT ANALYSIS MODEL TO EVALUATE PLAYER CHAT MESSAGES IN STARCRAFT 2. THEIR STUDY AIMED TO UNDERSTAND THE EMOTIONAL TONE OF PLAYER COMMUNICATIONS AND ITS IMPACT ON GAMEPLAY. BY APPLYING A SENTIMENT ANALYSIS MODEL, THE RESEARCHERS WERE ABLE TO CLASSIFY CHAT MESSAGES AND ASSESS THE PREVALENCE OF POSITIVE AND NEGATIVE SENTIMENTS DURING MATCHES. THIS ANALYSIS PROVIDED INSIGHTS INTO PLAYER BEHAVIOR AND THE OVERALL ATMOSPHERE OF THE GAMING COMMUNITY.

LITERATURE REVIEW



“ PREVIOUS STUDIES SHOW THAT TAILORED LEXICONS AND MACHINE LEARNING MODELS IMPROVE TOXICITY DETECTION IN GAME CHATS. RESEARCH HAS ADDRESSED CONTEXT-SPECIFIC LANGUAGE AND GENDER-BASED TOXICITY. THIS PROJECT BUILDS ON THESE FINDINGS USING A BERT MODEL TO CLASSIFY VALORANT AND MINECRAFT CHATS, ENHANCING MODERATION AND INCLUSIVITY “

SENTIMENT ANALYSIS IN WORLD OF TANKS

MURNION ET AL. (2018) EXPLORE THE APPLICATION OF MACHINE LEARNING AND SEMANTIC ANALYSIS IN DETECTING CYBERBULLYING WITHIN IN-GAME CHAT INTERACTIONS. THEIR STUDY EMPLOYS NATURAL LANGUAGE PROCESSING (NLP) TECHNIQUES TO ANALYZE TEXTUAL DATA, IDENTIFYING PATTERNS ASSOCIATED WITH HARMFUL, OFFENSIVE, OR TOXIC BEHAVIOR. BY LEVERAGING CLASSIFICATION ALGORITHMS AND LINGUISTIC FEATURE EXTRACTION, THE RESEARCHERS AIM TO ENHANCE THE EFFECTIVENESS OF AUTOMATED MODERATION SYSTEMS IN ONLINE GAMING ENVIRONMENTS, CONTRIBUTING TO THE BROADER FIELD OF DIGITAL SAFETY AND CONTENT MODERATION.



SENTIMENT ANALYSIS IN LEAGUE OF LEGENDS

PONGKHAN ET AL. (2024) INVESTIGATE SENTIMENT ANALYSIS IN THE LEAGUE OF LEGENDS GAMING COMMUNITY, FOCUSING ON IN-GAME CHAT CONVERSATIONS. THEIR STUDY ADDRESSES CHALLENGES IN PROCESSING UNSTRUCTURED TEXT RICH IN GAMING SLANG, ABBREVIATIONS, AND UNCONVENTIONAL GRAMMAR. BY LEVERAGING NATURAL LANGUAGE PROCESSING (NLP) TECHNIQUES, THEY ANALYZE PLAYER SENTIMENTS TO BETTER UNDERSTAND COMMUNITY DYNAMICS AND IDENTIFY TOXIC BEHAVIOR, CONTRIBUTING TO ENHANCED MODERATION AND PLAYER EXPERIENCE IN ONLINE GAMING ENVIRONMENTS.

DATA COLLECTION



OBJECTIVE

TO COLLECT IN-GAME CHAT MESSAGES FROM VALORANT AND MINECRAFT FOR TOXICITY CLASSIFICATION.

DATA SOURCES

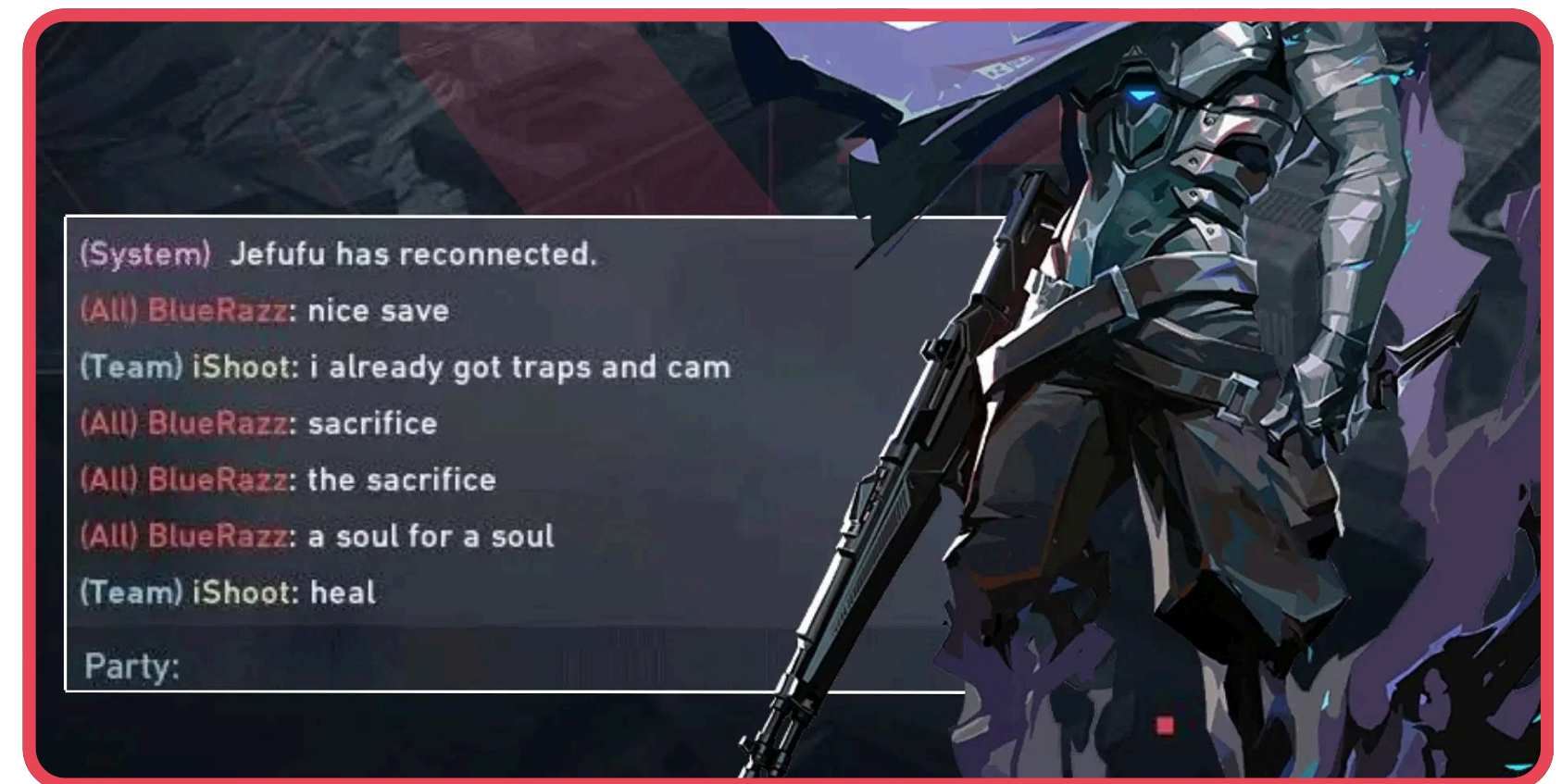
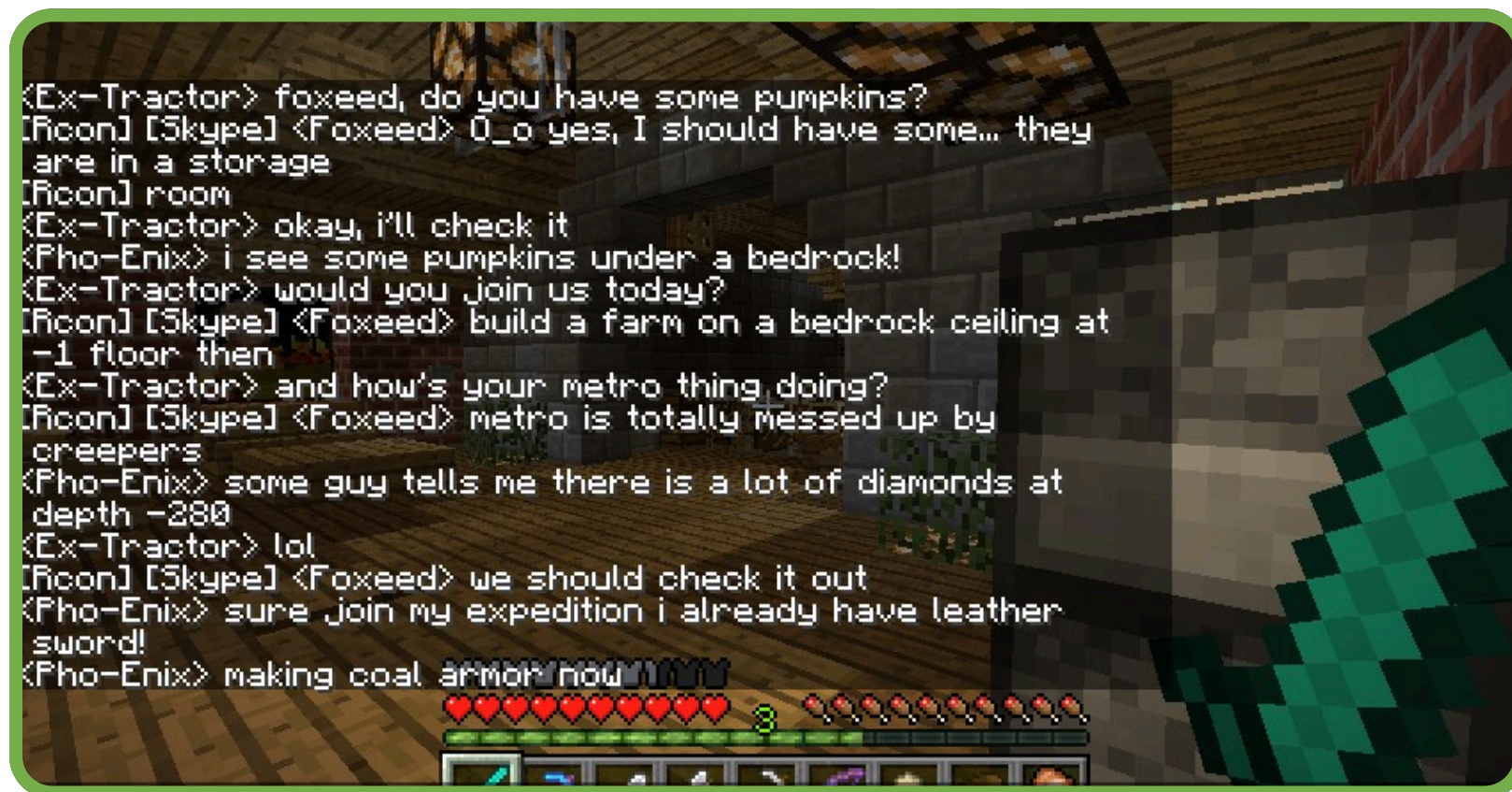
VALORANT: SCREENSHOTS OF IN-GAME CHAT LOGS DURING GAMEPLAY AND LIVESTREAMS.

MINECRAFT: SCREENSHOTS OF CHAT MESSAGES FROM PUBLIC SERVERS AND LIVESTREAMS.

DATA VOLUME

COLLECTED 1781 CHAT SCREENSHOTS.

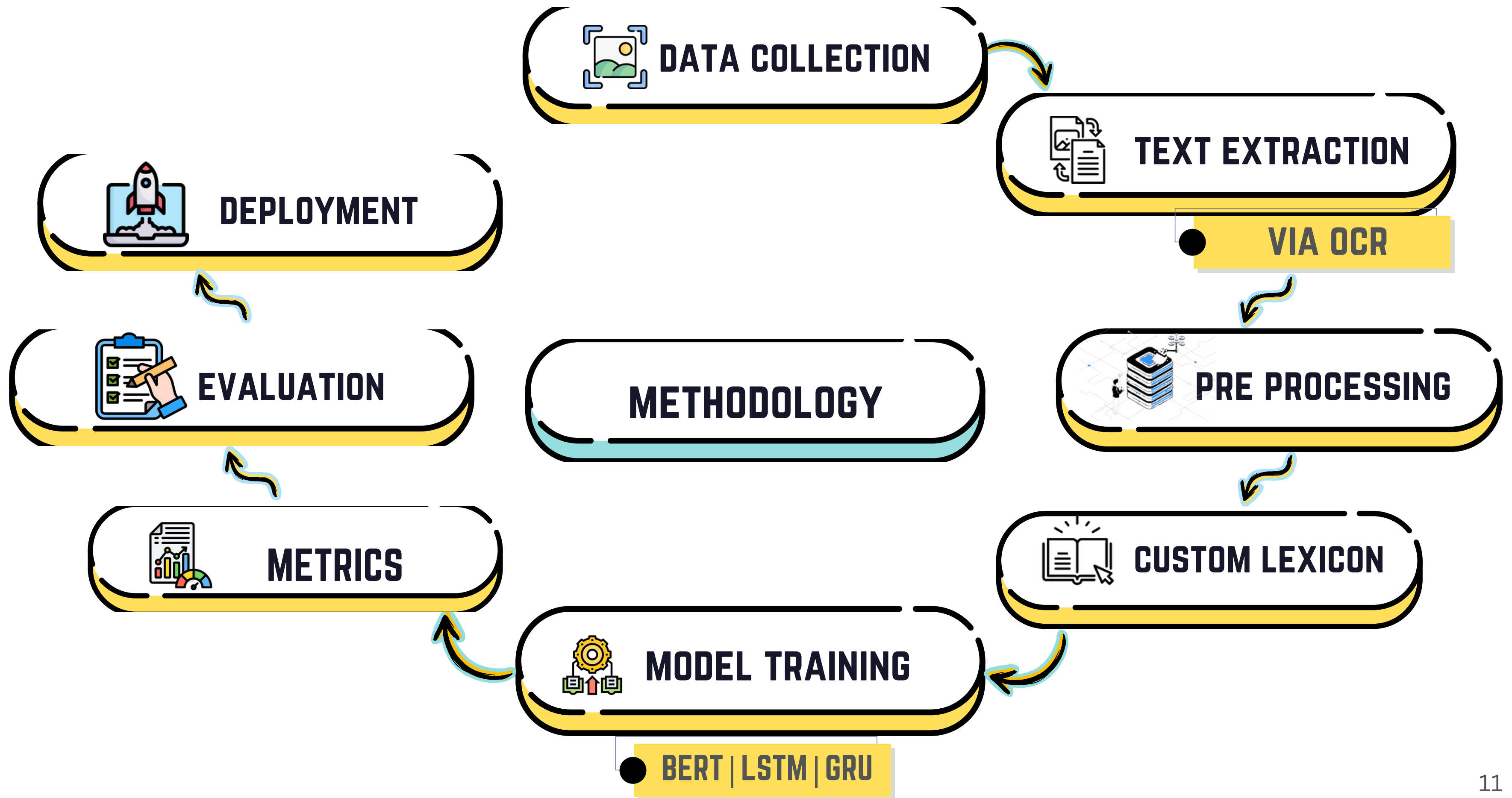
MINECRAFT x VALORANT



DATA COLLECTION PROCESS

CAPTURED HIGH-QUALITY SCREENSHOTS OF IN-GAME CHATS.

USED IN-GAME SCREENSHOT FEATURES.



TEXT EXTRACTION



- EXTRACTED TEXT FROM IMAGES USING PREPROCESSING (GRAYSCALE CONVERSION, RESIZING, SHARPENING) AND TESSERACT OCR. PROCESSED IMAGES FROM A DIRECTORY, APPLIED PREPROCESSING, AND USED PYTESSERACT FOR TEXT EXTRACTION.
- KEY LIBRARIES: OPENCV FOR IMAGE PROCESSING AND PYTESSERACT FOR OCR.

PREPROCESSING



- **PREPROCESSED TEXT DATA BY CONVERTING TO LOWERCASE, REMOVING SPECIAL CHARACTERS/NUMBERS, TOKENIZING, REMOVING STOPWORDS, AND LEMMATIZING USING NLTK.**
- **KEY LIBRARIES: PANDAS FOR DATA HANDLING, NLTK FOR TEXT PROCESSING, AND REGEX FOR TEXT CLEANING.**

CUSTOM LEXICON



- **CREATED A CUSTOM LEXICON TO CLASSIFY TEXT INTO TOXIC AND NEUTRAL CATEGORIES BASED ON PREDEFINED KEYWORDS AND WEIGHTS.**
- **DEFINED TOXIC KEYWORDS (E.G., "NOOB", "TRASH", "STUPID") AND NEUTRAL KEYWORDS (E.G., "GG", "WP", "ROTATE") WITH ASSOCIATED SCORES.**
- **CLASSIFIED PREPROCESSED TEXT BY CALCULATING SCORES FOR TOXIC AND NEUTRAL KEYWORDS USING THE CUSTOM LEXICON. ASSIGNED LABELS (TOXIC OR NEUTRAL) BASED ON THE HIGHER SCORE; DEFAULTED TO NEUTRAL IF SCORES WERE EQUAL OR ABSENT.**

VISUALIZATION



Category Distribution

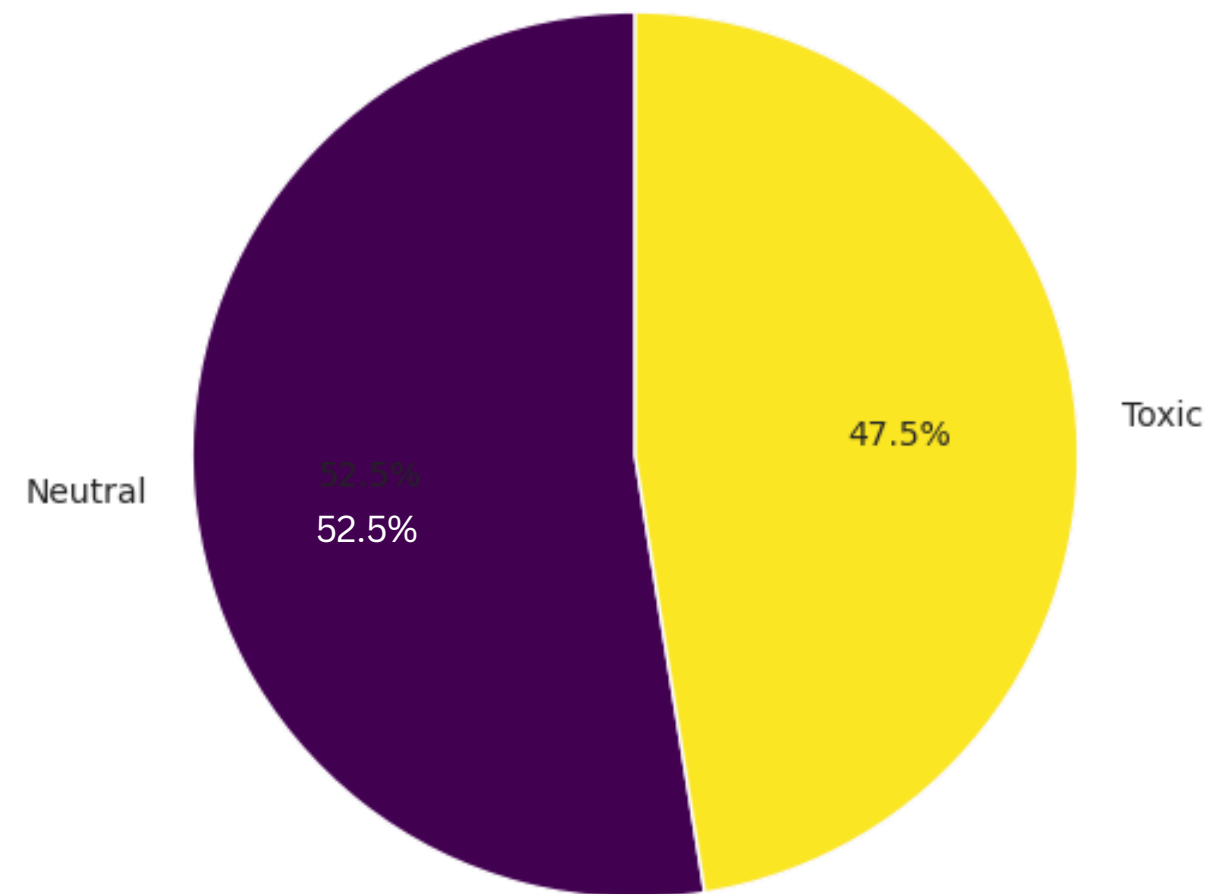


FIG 1. PIE CHART DEPICTING A
BALANCED DATASET

Most Common Words in Sentences



FIG 2. WORD CLOUD EXPRESSING
MOST FREQUENT USED WORDS

VISUALIZATION

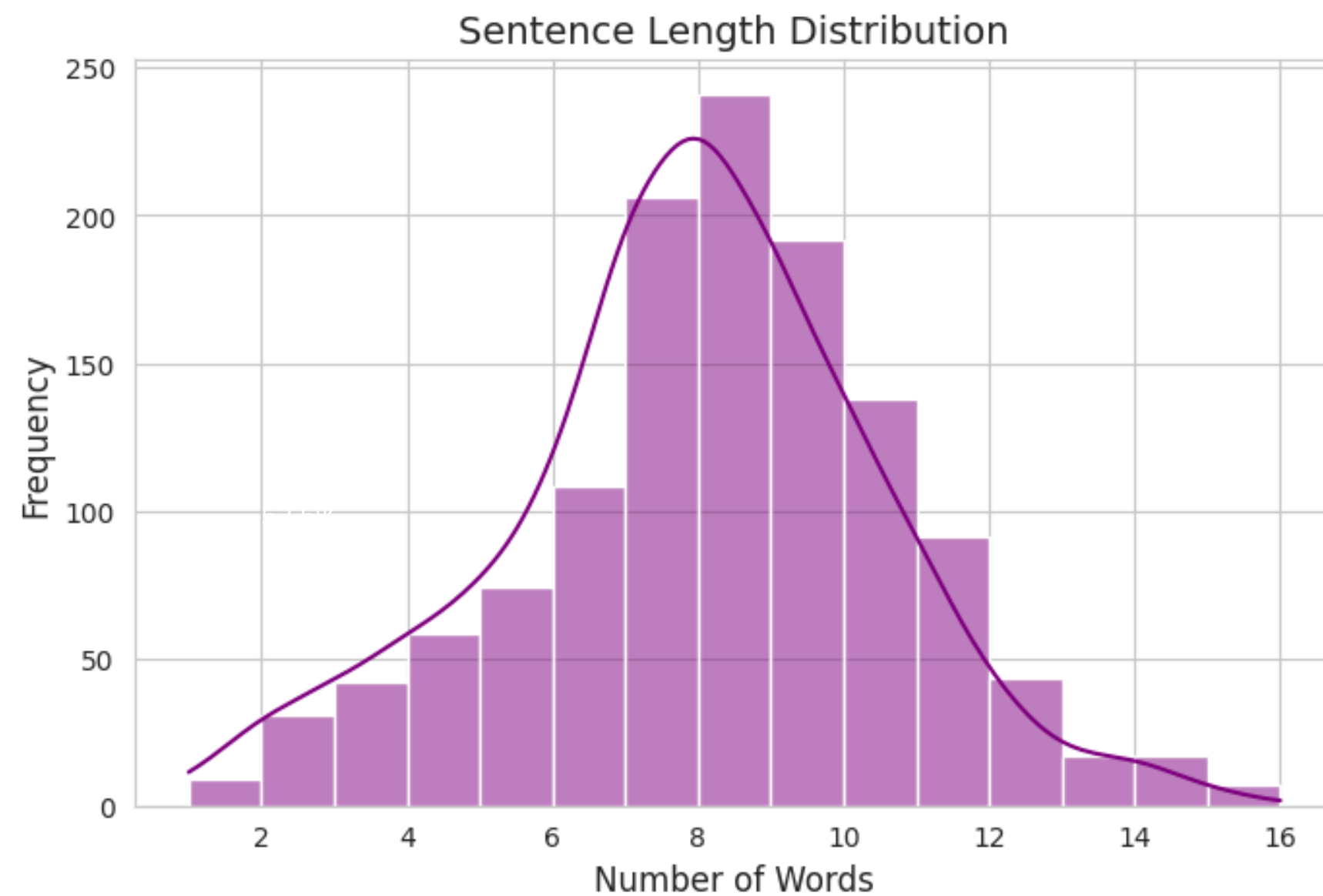


FIG 3. HISTOGRAM SHOWING THE AVERAGE LENGTH OF CHATS

MODEL TRAINING



- **TRAINED THREE MODELS** (BERT, LSTM, AND GRU) FOR TOXICITY CLASSIFICATION USING PREPROCESSED TEXT DATA
 - ✓ BERT – PRETRAINED TRANSFORMER WITH CUSTOM CLASSIFICATION HEAD.
 - ✗ LSTM – BIDIRECTIONAL LSTM WITH EMBEDDING LAYERS.
 - ✗ GRU – GRU-BASED MODEL FOR SEQUENCE LEARNING.
- **TRAINING PROCESS:**
 - DATA SPLIT: TRAINING & VALIDATION.
 - USED CROSSENTROPYLOSS AND ADAMW OPTIMIZER.
 - BATCH PROCESSING VIA PYTORCH'S DATALOADER.
- **SAVED TRAINED MODELS FOR DEPLOYMENT:**
 - BERT_TOXICITY_CLASSIFIER.PTH
 - LSTM_TOXICITY_CLASSIFIER.PTH
 - GRU_TOXICITY_CLASSIFIER.PTH

METRICS USED



- **TO ASSESS THE PERFORMANCE OF BERT, LSTM, AND GRU MODELS FOR TOXICITY CLASSIFICATION USING LOSS AND ACCURACY AS THE EVALUATION METRIC.**
- **BERT'S SUPERIOR PERFORMANCE COMES FROM ITS STRONG CONTEXTUAL UNDERSTANDING. LSTM AND GRU, THOUGH SIMPLER, MAY BE USEFUL IN RESOURCE-LIMITED SCENARIOS.**
- **BERT WAS THE MOST EFFECTIVE FOR DETECTING TOXIC SPEECH. LSTM & GRU STRUGGLED, MAKING THEM LESS VIABLE FOR THIS TASK.**

EVALUATION



AFTER TRAINING AND TESTING, HERE'S HOW THE MODELS PERFORMED:

MODEL	ACCURACY	LOSS
BERT	89%	0.266
LSTM	52%	0.69
GRU	50.39%	0.6952

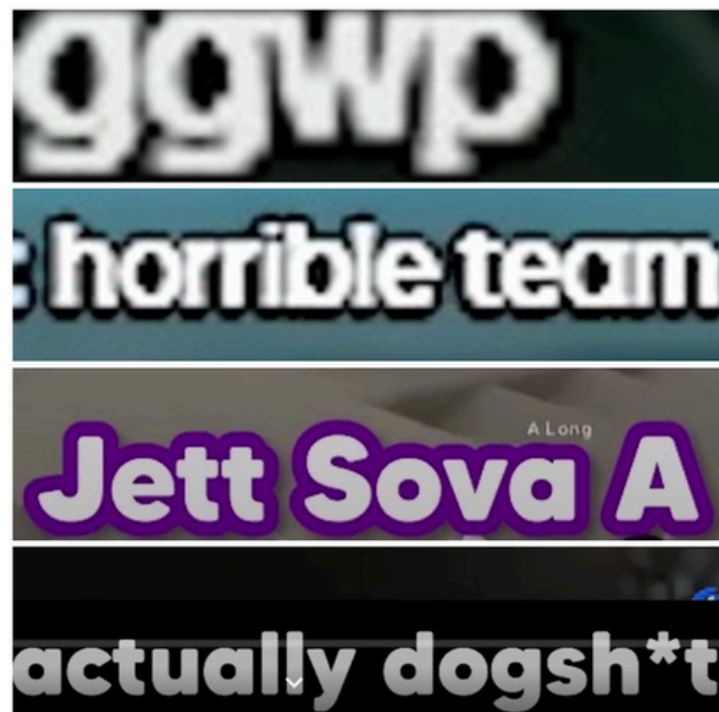
WHY DID BERT DOMINATE?

LSTM & GRU HOVERED AROUND 50%, MEANING THEY STRUGGLED TO LEARN TOXICITY PATTERNS EFFECTIVELY.

DEPLOYMENT



- USED TESSERACT AND OCR FOR TEXT EXTRACTION. CLEANED EXTRACTED TEXT BY REMOVING UNWANTED CHARACTERS AND FIXING COMMON OCR ERRORS.
- DEPLOYED A BERT-BASED TOXICITY CLASSIFICATION MODEL TO CLASSIFY TEXT EXTRACTED FROM UNSEEN IMAGES USING OCR (OPTICAL CHARACTER RECOGNITION).
- CLASSIFIED TEXT AS TOXIC OR NEUTRAL USING THE TRAINED BERT MODEL, ACHIEVING REAL-TIME PREDICTIONS.
- **EXAMPLE OUTPUTS:** "OUTSIDE A MAIN" → NEUTRAL, "IM ACTUALLY DOGSHT" → TOXIC



```
im actually dogsht -> Classification: Toxic  
Extracted Text: Oie Jett Sova A -> Classification: Neutral  
Extracted Text: horrible team -> Classification: Toxic  
Extracted Text: ggWpP -> Classification: Neutral
```

DEPLOYMENT



- THE SECOND APPROACH FOR VALIDATION WE TOOK IS REAL-TIME CHAT CLASSIFICATION.
- IN THIS, THE CHATS OF RECENTLY SELF-PLAYED GAMES WERE ASKED TO BE CLASSIFIED.
- THESE CHATS WERE LOADED ONTO OUR MOST OPTIMAL BERT MODEL AND ASKED TO CLASSIFY THESE CHATS
- THE MODEL WAS ABLE TO CLASSIFY THESE SENTENCES ACCURATELY, PROVING TO BE EFFICIENT AND ROBUST.

```
Sentence: Brainless coward -> Classification: Toxic
Sentence: GG boys -> Classification: Neutral
Sentence: Wp -> Classification: Neutral
Sentence: Too ezzz -> Classification: Toxic
Sentence: Can you pass me the cobblestone -> Classification: Neutral
Sentence: there's mlg clutch -> Classification: Neutral
Sentence: pass me the enchantment table -> Classification: Neutral
Sentence: Braindead person -> Classification: Toxic
Sentence: Dogsh*t -> Classification: Toxic
Sentence: Brainless coward -> Classification: Toxic
Sentence: Can you just learn how to play stupid? -> Classification: Toxic
```

RESULTS



KEY HIGHLIGHTS:

- SUCCESSFULLY CLASSIFIED TEXT FROM IMAGES AS TOXIC OR NEUTRAL USING A BERT-BASED MODEL.
- CUSTOM LEXICON IMPROVED DETECTION, ESPECIALLY FOR GAMING-SPECIFIC SLANG.
- PERFORMED DEPLOYMENT IN TWO WAYS TO ENSURE THE ACCURACY AND REAL-LIFE IMPLEMENTATION OF MODEL.

MODEL	ACCURACY	LOSS	
BERT	89%	0.266	✓
LSTM	52%	0.69	✗
GRU	50.39%	0.6952	✗

CONCLUSION & FUTURE SCOPE



SUMMARY :

- BERT IS THE BEST CHOICE FOR GAME CHAT TOXICITY DETECTION, ACHIEVING 89% ACCURACY AND MINIMUM LOSS OF 0.266.
- LSTM & GRU STRUGGLED, WITH ACCURACY CLOSE TO RANDOM GUESSING (~50%), MAKING THEM UNSUITABLE FOR THIS TASK.
- CUSTOM LEXICON IMPROVED DETECTION FOR GAME-SPECIFIC TERMS.
- PERFORMED VALIDATION IN TWO WAYS, ACHIEVING EXCELLENT RESULTS PROVING OUR MODEL'S EFFICIENCY AND ROBUSTNESS.

WHAT NEXT ?

- EXPAND THE DATASET TO COVER MORE DIVERSE IN-GAME CONVERSATIONS.
- DEVELOP A MULTI-LINGUISTIC MODEL TO ANALYZE TEXT FOR VARIOUS LANGUAGES.
- IMPLEMENTING SELF-LEARNING MODELS, AS WELL AS FRESH SLANG AND CERTAIN GAMING TRENDS, MAY POSITIVELY IMPROVE OVERALL LONG-TERM EFFECTIVENESS.
- REAL-TIME CHAT DATA GETS CREATED QUITE OFTEN BY GAMING PLATFORMS. OPTIMIZING THE MODEL WILL BE IMPORTANT FOR ITS SCALABILITY.



THANK YOU



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